

Elbow Imaging

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The elbow is a complex joint affected by a variety of pathologic conditions. Advanced imaging of the elbow with computed tomography (CT), magnetic resonance imaging (MRI), MR arthrography, and ultrasound (US) provide powerful tools for diagnosis; however, an understanding of the utility and limitations is required to select the optimal modality to answer each clinical question. Despite the availability of advanced imaging, conventional radiographs remain central to imaging of the elbow. It is important not to overlook conventional radiographs in the evaluation of the elbow; they are recommended in all patients prior to the consideration of advanced imaging techniques. This chapter reviews these imaging modalities and also considers how these modalities can best be utilized in the evaluation and treatment planning of commonly encountered elbow pathology.

IMAGING MODALITIES

Conventional Radiographs

Imaging evaluation of the elbow typically begins with conventional radiographs. Standard radiographic views of the elbow include anteroposterior (AP), lateral, and oblique projections (Fig. 59.1). The AP and oblique views are obtained with the elbow extended and forearm supinated, permitting visualization of the proximal radioulnar and radiocapitellar articulations, medial and lateral epicondyles, and trochlea. The carrying angle of the elbow typically measures 12 to 15 degrees valgus in this anatomic resting position.¹ The lateral view, obtained with the elbow flexed 90 degrees and the forearm in a neutral position, permits evaluation of the radiocapitellar and ulnotrochlear articulations, distal humerus, olecranon, and coronoid processes. An appropriately acquired lateral elbow radiograph clearly shows the ulnotrochlear articulation with minimal overlap of osseous structures. This ulnotrochlear space on the lateral view can be obscured with small amounts of rotation; therefore careful attention should be paid to obtaining well-positioned lateral elbow views. An anatomically positioned radial head is collinear with the capitellum. A dark radiolucent area is normally seen just anterior to the distal humerus, representing the anterior fat pad. Similarly, the posterior fat pad sign is a well-demarcated lucency posterior to the distal humeral metaphysis; its presence tends to indicate a pathologic process.² Displacement of these fat pads, produced by an intra-articular effusion or hemarthrosis, resembles the spinnaker on a sailboat and is referred to as the *sail sign*.³

Conventional radiographic projections are usually satisfactory for the initial evaluation of the elbow; however, additional projections may be utilized for structures that are difficult to evaluate on standard AP, lateral, and oblique projections. Fractures of the radial head are common and may be challenging to detect when nondisplaced. A specific radial head (radiocapitellar) view may better demonstrate the fracture, characterize the fracture pattern, and aid in classification. With the elbow flexed to 90 degrees, the beam is angled 45 degrees anteriorly from a true lateral.¹ This projection provides an unobstructed view of the radial head without overlap of the proximal ulna. Other common fractures include those involving the epicondyles (especially the medial epicondyle in skeletally immature patients), distal humerus, and olecranon. The “terrible triad” injury includes fractures of the radial head and coronoid process, along with dislocation of the elbow joint. Conventional radiographs are usually sufficient for confirming concentric reduction of the elbow after closed manipulation.

Any patient presenting with non-activity-related pain or a mass at the elbow should first be evaluated with conventional radiographs. Radiographs offer a large amount of information and help guide further imaging workup.

Elbow trauma often involves a combination of osseous and soft tissue injury. When conventional radiographs do not reveal a problem or provide incomplete information, the choice of additional studies is guided by the history and physical examination. These additional studies may include CT, MRI, MR arthrography, fluoroscopy, and bone scan. Some relevant injuries and their imaging findings are highlighted in this chapter to aid the general orthopedic surgeon or sports medicine specialist.

Computed Tomography

Modern CT scanners allow for rapid image acquisition, high spatial resolution, and isovolumetric data sets that can provide multiplanar reformatting of the images. These qualities allow for excellent depiction of bone; they are useful for the detection of subtle fractures, periosteal reaction, and subtle bone lesions that may be difficult to appreciate on conventional radiographs. The high attenuation of cortical bone allows for segmentation and generation of three-dimensional volume-rendered images with minimal manual postprocessing. These images provide important visual information in the planning of complex fracture repairs. CT technology continues to improve, with dose reduction, metal reduction techniques, and quantitative technology.

Abstract

The elbow is a complex joint commonly affected by a variety of pathologic conditions. Advanced imaging of the elbow—with computed tomography (CT), magnetic resonance imaging (MRI), MR arthrography, and ultrasound—provides powerful tools for diagnosis; however, an understanding of the utility and limitations is required to select the best optimal modality to answer each clinical question. Despite the availability of advanced imaging, conventional radiographs remain central to imaging of the elbow. It is important not to overlook conventional radiographs in the evaluation of the elbow; they are recommended in all patients prior to the consideration of advanced imaging techniques. This chapter reviews these imaging modalities and also considers how these modalities can best be utilized in the evaluation and treatment planning of commonly encountered elbow pathology.

Keywords

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CT
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UCL
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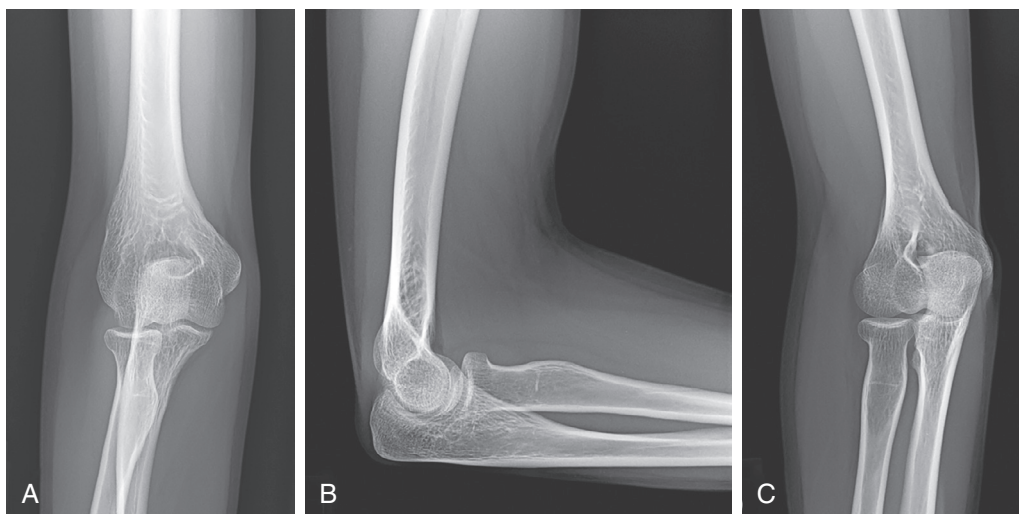


Fig. 59.1 Conventional radiographs of the elbow. (A) Anteroposterior view. (B) Lateral view. (C) Oblique view.

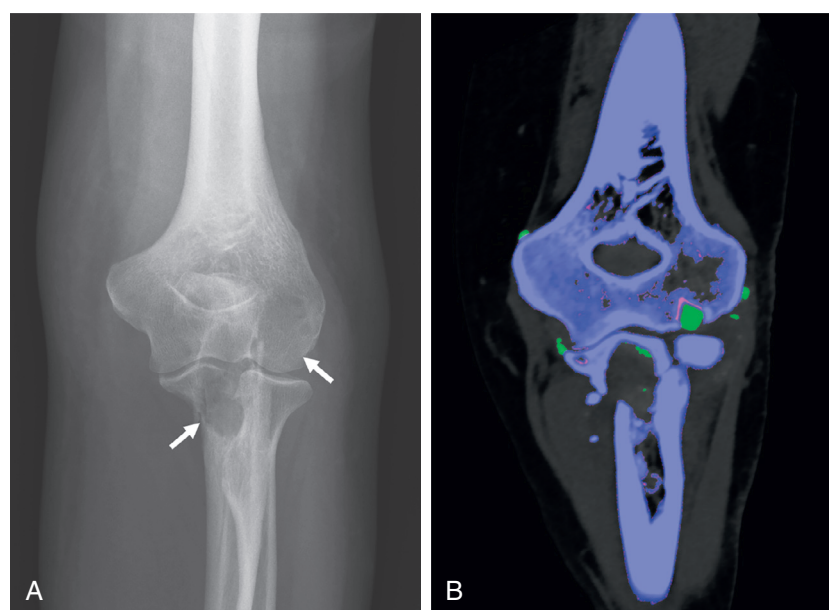


Fig. 59.2 Anteroposterior radiograph of the elbow (A) demonstrates lytic lesions in the proximal ulna and capitellum (arrows). Biopsy of the lytic ulnar lesion was negative for malignancy and demonstrated inflammation. A subsequent dual-energy computed tomography exam (B) confirmed the presence of a large amount of uric acid crystal deposition in the elbow as indicated by the green encoded pixels on the postprocessed coronal image, confirming the diagnosis of gout.

For example, dual-energy CT (DECT) can be used in the musculoskeletal system for the detection of uric acid crystals in gout.⁴ The foot and ankle are most commonly imaged with this DECT technique, but it can also be applied to the elbow in challenging diagnostic cases (Fig. 59.2).

Magnetic Resonance Imaging

MRI provides excellent soft tissue contrast and is the modality of choice for evaluating soft tissue masses, ligament injury, tendon injury, nerve-related symptoms, and articular cartilage. It is also the modality of choice in evaluating clinically relevant nonspecific elbow pain not explained by conventional radiographs. There is no single MRI protocol for the elbow, and the MRI technique

should be tailored to the specific clinical question in order to provide the best diagnostic information. T1-weighted, proton density and fluid-sensitive sequences are typically adequate for evaluating the elbow. T1-weighted images better depict osseous pathology and are preferred in the evaluation of a soft tissue mass and of the nerves. Ligaments, tendons, and cartilage are better depicted with proton density sequences. The term *fluid sensitive* refers to an MRI sequence that depicts the protons in fluid/edema as bright while the protons in fat appear dark. Examples of fluid-sensitive sequences include T2-weighted images with chemical fat saturation, intermediate proton density-weighted with chemical fat saturation, short tau inversion recovery (STIR), and Dixon water/fat separation. Intravenous gadolinium

contrast is not typically required in elbow MRI but is often used for the evaluation of soft tissue masses, nerve pathology, or synovitis.

Magnetic Resonance Arthrography

MR arthrography may improve the MRI evaluation of locking, limited range of motion (ROM), and ulnar collateral ligament (UCL) injury, but preference between standard MRI and MR arthrogram for these diagnoses varies. The instillation of dilute gadolinium contrast into the elbow may be guided by US or fluoroscopy, depending on availability and the comfort of the proceduralist. The patient is most commonly positioned prone with the elbow above the head, but the injection can also be administered with the patient seated adjacent to the fluoroscopy table and the arm placed on the table. With the elbow flexed at 90 degrees, the elbow is positioned as in a lateral radiograph. A 22-gauge needle is placed in the radiocapitellar joint and a small test of iodinated contrast is used to confirm intra-articular placement. The dilute gadolinium contrast is instilled under intermittent fluoroscopic guidance to ensure intra-articular placement. The MR arthrography sequences differ, as the exam relies heavily on T1-weighted fat-saturated images. This allows for the detection of any gadolinium contrast that may extend into or through a ligament tear, as in the evaluation of an UCL tear.

Ultrasound

US excels at dynamic soft tissue evaluation of the elbow joint and is able to provide an alternative for imaging around surgical hardware. Because of the complex mechanics of hinging and rotation, static imaging may fail to detect dynamic pathology. US has the benefit of high-resolution dynamic image evaluation without ionizing radiation and allows for direct patient communication and “real-time” examination. US is a powerful tool but is user-dependent, requiring a detailed anatomic knowledge and basic understanding of sonographic principles if the operator is to select the correct transducer and be aware of imaging artifacts and pitfalls. Image interpretation can be optimized by using high-frequency linear array transducers (10 to 20 MHz), utilizing standardized imaging acquisition planes and acquiring stored cine clips for subsequent review.

The use of US in the evaluation of the elbow is growing. Currently it is most often used in the evaluation of dynamic pain/snapping, tendon pathology, joint effusion, peripheral nerve evaluation, and for guiding percutaneous procedures about the elbow.

TRAUMA

Fracture

The elbow features complex three-dimensional anatomy and overlapping structures that can make the detection of subtle fractures on conventional radiographs challenging. Interpretation of radiographs of the skeletally immature elbow for trauma requires knowledge of the ossification centers, the time line for development of the ossification centers, and an understanding of the order of growth plate closure. The mnemonic CRITOE can serve as an aid for remembering the order of ossification

and closure (C, capitellum; R, radial head; I, internal/medial epicondyle; T, trochlea; O, olecranon; E, external/lateral epicondyle). Avulsion fractures are common injuries of the pediatric elbow, and knowledge of this pattern of growth plate closure helps in their identification.

CT or MRI can help to detect subtle fractures. CT is often preferred, given its availability and lower cost; however, MRI is also excellent for detecting nondisplaced fractures. CT with multiplanar reformatted images is the preferred modality in the presurgical planning of complex elbow fractures.⁵ Therefore complex fractures of the distal humerus and medial coronoid facet are commonly evaluated by CT for preoperative planning.

Fracture-Dislocation

Elbow dislocations are evident on conventional radiographs and are characterized by the position of the radius/ulna relative to the humerus. Typically these injuries are not subtle on radiographs, but the associated deformity and difficulty of positioning can limit the detection and characterization of fractures. Traction radiographs may help delineate the fracture pattern but may not be tolerated by all patients owing to the associated pain. CT better depicts complex fracture-dislocations and aids in fracture pattern characterization, degree of fragment displacement, articular surface involvement, and/or the presence of associated intra-articular fracture fragments, which may block concentric reduction and lead to persistent joint subluxation. Failure to identify fractures of stabilizing osseous structures such as the coronoid or radial head may lead to a poor functional outcome.⁶ Osseous three-dimensional reconstructions are commonly created from CT data for these complex fractures of the elbow to best demonstrate the relationship of the fracture fragments and for surgical planning. Finally, MRI is the best option for determining the pattern of soft tissue disruption in acutely unstable injuries or those that present with delayed complaints of subjective instability, which may require collateral ligament repair or reconstruction.

Occult Fracture

Persistent tenderness over a suspected injury site despite negative findings of conventional radiographs may be an indication for further evaluation in select cases. The posterior fat pad sign in the setting of trauma corresponds to an occult fracture in more than 75% of patients.^{7,8} The radial head is the most frequent site of an occult fracture.⁸ Fractures of the coracoid process are best visualized on well-positioned lateral radiographs; however, oblique views may also be helpful in detection and characterization. CT or MRI better detect non-displaced fractures and are indicated with high clinical suspicion of a fracture and negative radiographs. Fluid-sensitive (fat-suppressed T2-weighted or inversion recovery) sequences are the most sensitive for detecting radiographically occult fractures.⁷ Fractures show a linear pattern of signal change, with decreased signal on T1-weighted images and increased signal on T2-weighted images (Fig. 59.3A and B). Proton density images are commonly utilized in examining the elbow, but T1 images better demonstrate the linear signal abnormality of a fracture (see Fig. 59.3A). In contrast, osseous contusion produces a nonspecific diffuse increase in signal on T2-weighted images without a discrete fracture line (see Fig. 59.3C).



Fig. 59.3 Sagittal T1- (A) and T2-weighted fat-saturated (B) magnetic resonance images of the elbow after trauma demonstrate a radiographically occult acute fracture of the radial head and neck. Note the linear decreased signal abnormality of the fracture on both the T1- and T2-weighted sequences (*arrows*) and the surrounding bone marrow edema. Sagittal T2-weighted fat-saturated image of the radiocapitellar joint (C) in a different patient after trauma demonstrates the appearance of an osseous contusion with amorphous bone marrow edema in the absence of a discrete linear signal abnormality.

Stress Fracture

The olecranon is the most common stress fracture site, and this condition is most often seen in baseball players.⁹ When the history and physical examination suggest it, a CT, MRI, or bone scan may be diagnostic.¹⁰ A CT scan typically reveals a subtle fracture line in the olecranon, often with a linear area of sclerosis. Findings on MRI resemble those of occult fractures described previously. A bone scan would reveal abnormal radiotracer uptake at the site of repetitive stress.

DEGENERATIVE ARTHRITIS AND LOOSE BODIES

Pain and limited ROM associated with degenerative arthritis of the elbow can be quite limiting. Prior trauma may accelerate the development of degenerative arthritis, and surgical intervention may be beneficial in restoring function and reducing pain. The goal of imaging is to assist in determining potential pain-generating sites and/or a structural cause limiting ROM. Conventional radiographs may detect intra-articular loose bodies in the coronoid or olecranon fossae and osteophytes that may limit range of flexion or extension. Flexion and extension radiographs may be useful to illustrate how the structural abnormality is affecting the arc of motion. Supplemental oblique or axial projections may be helpful in distinguishing intra-articular and extra-articular calcifications, especially in severe cases of heterotopic ossification. In ambiguous cases, CT is the modality of choice for further information. Singson et al.¹¹ compared the utility of double-contrast CT arthrography and conventional radiography in patients with pain, locking, and limited elbow motion. They found that double-contrast CT arthrography successfully diagnosed 100% of loose bodies and provided precise information regarding the size, number, and location of the lesions. In contrast, conventional radiographs identified only 50% of the intra-articular loose bodies. Zubler et al.¹² arrived

at the same conclusion, noting greater accuracy of loose body detection with CT than with conventional radiographs, but particularly in the posterior fossae.

On the other hand, not all investigators agree that CT is necessary for loose body detection. Dubberley et al.¹³ found that CT and MRI were no more effective than conventional radiography alone for the detection of loose bodies. Quinn et al.¹⁴ recommended MRI for the accurate assessment of elbow intra-articular loose bodies. Current standard CT and MRI techniques are typically adequate for detecting intra-articular bodies without the need for arthrography. An advantage of using MRI is that it enables the detection of cartilaginous or osteocartilaginous fragments⁷ associated with osteochondritis dissecans of the capitellum that, in young athletes, may elude characterization by conventional radiography or CT. Furthermore, MRI may distinguish osteophytes and synovial hypertrophy, which often mimic loose body formation.¹⁵

COLLATERAL LIGAMENT INJURY

Conventional radiographs are usually insufficient for detecting collateral ligament injury of the elbow; however, soft tissue edema or hematoma can be identified. Ligamentous insufficiency may affect elbow alignment on standard radiographs. The normal alignment of the radiocapitellar joint can be altered on conventional radiographs with posterolateral rotatory instability, the most common pattern of elbow instability.¹⁶ Severe posterolateral rotatory instability may alter the radiocapitellar alignment on conventional radiographs with posterolateral subluxation of the radial head. Collateral ligament instabilities of the elbow can be evaluated with stress imaging, including radiographs or US. Elbow instability resulting from collateral ligament injury or varus/valgus stress is applied to visualize asymmetric widening of the joint.^{17,18} US has the benefit of dynamic visualization and the ability to compare an image with one of the contralateral elbow (Fig. 59.4). Other examinations—such as CT arthrography, MRI, and MR arthrography—also offer additional information by direct visualization of the ligament.

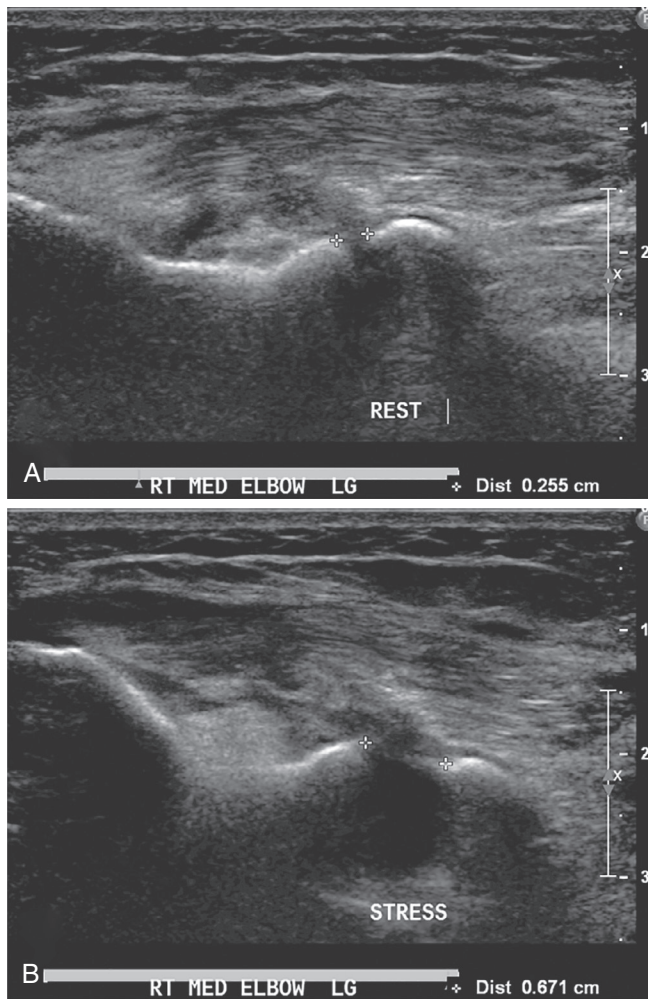


Fig. 59.4 Longitudinal ultrasound images of the medial right elbow were performed at rest (A) and with valgus stress (B). This exam demonstrates 4 mm of medial joint space widening with valgus stress in a patient with a complete ulnar collateral ligament tear.

Ulnar Collateral Ligament Injury

The UCL is an important valgus stabilizer of the elbow and is particularly vulnerable to injury in athletes whose sport involves throwing, such as baseball pitchers and javelin throwers.¹⁹ Interpretation of valgus stress radiography in athletes can be challenging. Several investigators have shown increased valgus laxity in the dominant arm of asymptomatic persons.^{17,20,21} When the index of clinical suspicion for a UCL injury is high, MRI provides important additional information. The UCL is a vertically oriented structure coursing between the medial epicondyle and the coronoid process that is uniformly of low signal intensity; it is best depicted on coronal MRI. The UCL is composed of anterior, posterior, and transverse bands. The anterior bundle provides the most valgus elbow stabilization. MRI may detect full-thickness tears of the anterior bundle, with increased signal intensity often found within and adjacent to the discontinuous ligament on fat-suppressed T2-weighted images as a result of edema and/or hemorrhage (Fig. 59.5A).^{7,22} MRI is less reliable for the detection of partial-thickness tears. Timmerman et al.²³ found MRI to be 100% sensitive for full-thickness tears but only 14% sensitive for partial-thickness tears. Administration of intra-articular contrast material improves sensitivity^{7,22–24} in detecting partial-thickness UCL tears (see Fig. 59.5B). Schwartz et al.²⁵ reported 86% and 100% sensitivity and specificity, respectively, for the detection of partial-thickness UCL tears with MR arthrography. Variability in the insertion of the anterior bundle, however, complicates MRI interpretation of partial-thickness tears. The distal ulnar attachment of the anterior bundle may insert anywhere between 1 mm of the articular margin of the coronoid process²⁵ and 3 mm distal to the sublime tubercle of the ulna.²⁶ This characteristic can create a small recess on MR arthrography along the medial margin of the coronoid process. Partial tears of the distal insertion of the UCL are diagnosed by intra-articular gadolinium extending deep to the insertion, referred to as the

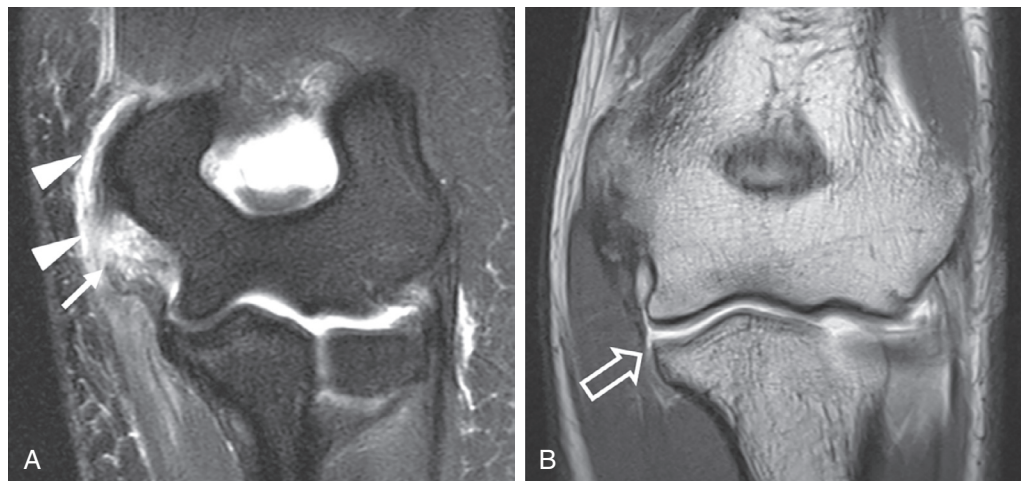


Fig. 59.5 Tears of the ulnar collateral ligament (UCL). (A) Coronal T2-fat-saturated magnetic resonance image demonstrates an acute full-thickness tear of the proximal UCL of the elbow (arrow) and surrounding edema (arrowheads). (B) Coronal T1-weighted elbow arthrogram shows a pitcher with a partial tear of the distal UCL at the undersurface of the insertion at the sublime tubercle (open arrow). This partial undersurface tear of the distal UCL is referred to as the “T-sign,” given the appearance of a horizontal T by the intra-articular gadolinium contrast.

“T-sign” (see Fig. 59.5B). Consequently distinguishing between normal anatomy and a pathologic partial undersurface tear at the attachment of the distal ligament can be challenging²⁷; therefore clinical correlation is paramount.

US examination of the UCL can aid in the diagnosis of a tear and can evaluate for instability with valgus stress. The UCL is best depicted at 70 degrees of flexion with the transducer placed in a longitudinal manner to the long axis of the ligament (Fig. 59.6).²⁸ In this position, the anterior band of the UCL should appear taut, with a tear demonstrated as focal hypoechoogenicity with fiber disruption.²⁹ The medial joint margin and UCL can be dynamically evaluated under valgus stress for determining laxity/insufficiency, with greater than 2 mm of widening compared with the contralateral elbow being considered an abnormality of the anterior band of the UCL.³⁰ The prevalence of UCL thickening, irregularity, and laxity complicates the US evaluation of the UCL in pitchers, and these abnormalities tend to progress over time.³¹

Injury of the Lateral Collateral Ligament

The lateral UCL (LUCL), radial collateral ligament (RCL), and annular ligament form the lateral collateral ligament complex. The LUCL is found posteriorly extending from the lateral humeral

condyle to the supinator crest of the ulna. The RCL arises from the lateral humeral epicondyle, deep to the common extensor tendon, and blends into the annular ligament surrounding the radial head. On MRI, the RCL and LUCL are best seen on sequential coronal images with an image slice thickness of 2 mm or less.³² The annular ligament is optimally seen on axial MRI images.²⁷ The LUCL is the primary varus elbow stabilizer.

Tears of the lateral collateral ligament complex are common in persons with acute elbow dislocations. MRI findings of acute LUCL and RCL tears include ligament attenuation, redundancy, and discontinuity evident on coronal and/or axial images^{22,33}; they are most often seen at the humeral origin of the ligament (Fig. 59.7).³² Chronic insufficiency of the LUCL, which can lead to posterolateral rotatory instability, is difficult to identify on MRI.^{34,35} Terada et al.³⁶ found that MRI of asymptomatic persons revealed inconsistent signal characteristics within the LUCL or frank inability to identify the structure altogether. In contrast, Potter et al.³⁷ found that abnormalities in the LUCL could be reliably detected with three-dimensional gradient-recalled and fast spin echo MRI in persons with posterolateral rotatory instability compared with asymptomatic control subjects.³⁷ The role of MR arthrography in this setting is unclear; it is not currently deemed to be a superior method of characterizing injuries of this ligament complex.

The RCL originates intimately with the origin of the common extensor tendon. The separate RCL structure can be delineated from the common extensor tendon origin with knowledge of the humeral footprint.³⁸ The LUCL originates with and cannot be readily distinguished from the RCL proximally. The oblique course of the middle and distal LUCL makes US evaluation particularly challenging.³⁹ Posterolateral ulnohumeral laxity can be quantified by using dynamic US in patients with suspected posterolateral rotatory instability.⁴⁰

LATERAL AND MEDIAL EPICONDYLITIS

Lateral epicondylitis, or “tennis elbow,” is the most common overuse disorder of the elbow. Conventional radiographs rarely show any findings in persons with lateral epicondylitis. Pomerance⁴¹ found radiographic abnormalities in only 16% of 294 patients with lateral epicondylitis; the most common manifestation was faint calcific deposits at the origin of the extensor carpi radialis brevis (Fig. 59.8A). MRI has 90% to 100% sensitivity and 83% to 100% specificity for the detection of lateral epicondylitis.⁴² However, it is reserved for equivocal or recalcitrant cases in order to exclude concomitant/alternative sources of elbow pain such as degenerative joint disease, osteochondral lesions, radial neuritis, posterolateral rotatory instability, occult fracture, and/or loose bodies.⁴³ Common MRI findings include high signal intensity on T2-weighted and STIR sequences in a thickened common extensor tendon with or without adjacent soft tissue edema (see Fig. 59.8B and C).^{22,33}

Medial epicondylitis, or “golfer’s elbow,” is the next most common overuse disorder of the elbow. Although conventional radiographs are usually unremarkable in middle-aged patients with degenerative tendinopathies, concomitant medial ulnotrochlear osteophytes may be present in this age group and thus have

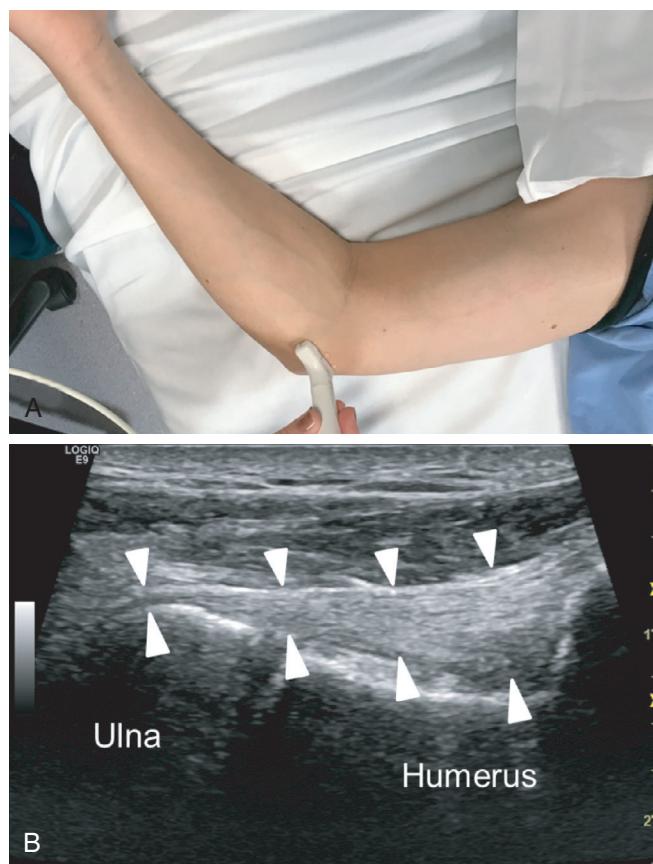


Fig. 59.6 Ultrasound (US) imaging of the ulnar collateral ligament (UCL). A photograph (A) demonstrates the US probe position for evaluating the UCL with the elbow in 70 degrees of flexion. The longitudinal US image (B) was obtained with the Logiq E9 US system (GE Healthcare, Milwaukee, WI) using a wide-band high-frequency probe (8 to 18 MHz). The image demonstrates the normal appearance of the UCL (arrowheads).

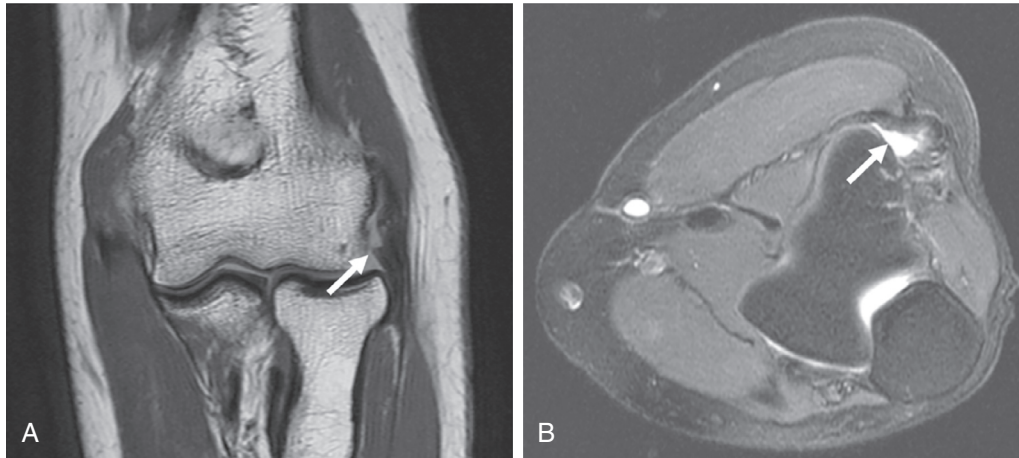


Fig. 59.7 Avulsion of the radial collateral ligament. (A) Coronal proton density weighted magnetic resonance image demonstrates a complete avulsion of the radial collateral ligament (*arrow*). (B) The axial T2-weighted fat-saturated image confirms complete avulsion of the origin of the radial collateral ligament (*arrow*). Note that the overlying common extensor tendon remains intact. Exam under anesthesia was positive for posterolateral rotatory instability and followed by surgical confirmation and reconstruction.

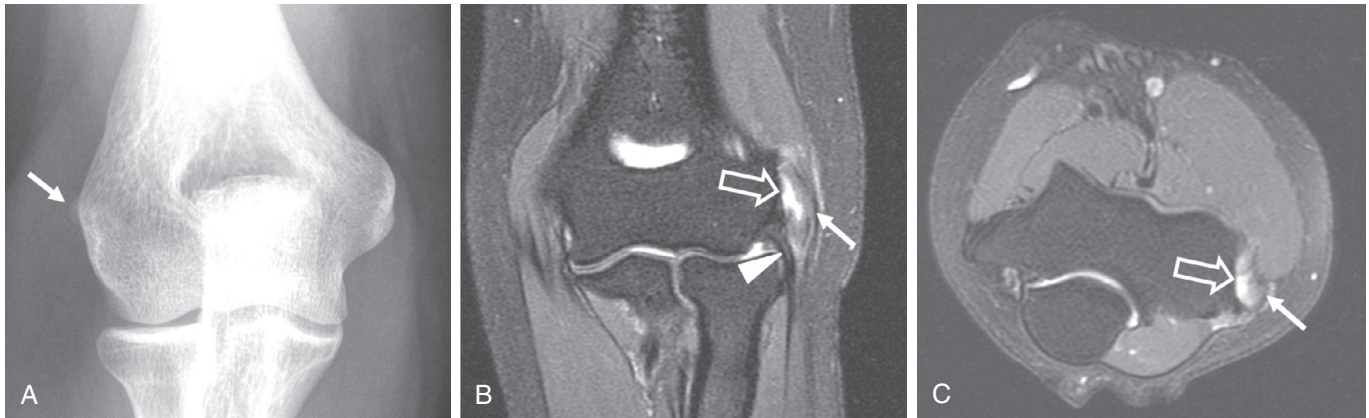


Fig. 59.8 Lateral epicondylitis. (A) Anteroposterior radiographs of the elbow demonstrates faint heterotopic ossification at the origin of the common extensor tendon (*arrow*). (B) Coronal and (C) axial T2-weighted fat-saturated magnetic resonance imaging (MRI) illustrates the MRI findings in lateral epicondylitis. The common origin of the extensor tendon demonstrates marked thickening and increased intrasubstance signal (*arrows*). A partial tear at the origin of the common extensor tendon is seen as a cleft of fluid (*open arrows*). Note the intact underlying radial collateral ligament (*arrowhead*).

implications for the overall treatment plan. MRI is beneficial only for cases of refractory medial epicondylitis, mostly to exclude other sources of pain such as UCL injury or ulnar neuritis.²⁷ A thickened common flexor tendon may be seen on T1-weighted sequences, with local soft tissue edema surrounding the medial epicondyle on T2-weighted sequences. Degenerative tendinosis is typically localized to the origins of the flexor carpi radialis and pronator teres.²⁷

Abnormal tendon hypoechogenicity, thickening and decreased fibrillary echotexture are the US features of tendinopathy seen in both medial and lateral epicondylitis. A tear appears as a linear anechoic focus with associated fiber disruption.⁴⁴

Distal Biceps Tendon

Diagnosis of distal biceps tendon rupture by conventional radiography is limited because osseous avulsion injuries from the

insertion of the radial tuberosity are rare. MRI may distinguish degenerative tendinosis and acute injury and differentiate partial from complete tears of the tendon. Fitzgerald et al.⁴⁵ reported a 100% accuracy rate of MRI for the diagnosis of distal ruptures of the biceps tendon. A complete rupture is characterized by an interruption in the low-signal-intensity tendon insertion at the radial tuberosity on axial images combined with an associated large amount of high-signal-intensity fluid and/or hemorrhage in the antecubital fossa seen on T2-weighted images (Fig. 59.9). The degree of tendon retraction is best assessed on sagittal images. Partial tears or nonretracted distal avulsions can be challenging to diagnose and differentiate on standard MRI planes of the elbow. The distal biceps tendon courses to the medial aspect of the radius to insert on the radial tuberosity, which may limit the evaluation of insertional injury. The radial insertion of the biceps is well demonstrated with the elbow flexed 90 degrees,

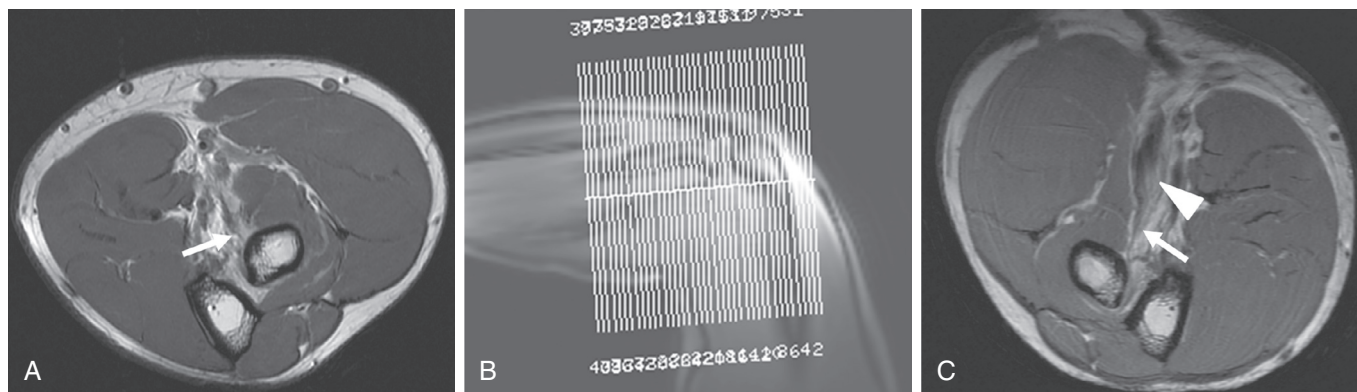


Fig. 59.9 Distal biceps tendon avulsion. (A) Axial proton-density image of the elbow demonstrates a complete avulsion of the distal biceps tendon (*arrow*) at the insertion on the radial tuberosity. To better define the tear and the quality of the distal biceps tendon, the patient was placed in the FABS position (elbow flexion, shoulder abduction, and forearm supination). (B) Scout image in the FABS position shows the plane of acquisition perpendicular to the long axis of the proximal radius. (C) Proton density axial FABS image shows the distal biceps avulsion (*arrow*) and the thickening and increased intrasubstance signal in the biceps tendon stump (*arrowhead*).

the shoulder abducted, and the forearm supinated. This position is commonly referred to as the FABS position, for Flexed elbow, shoulder ABduction, and Supination of the forearm. Images are acquired perpendicular to the long axis of the proximal radial shaft in the FABS position, allowing for a long segment of the distal biceps tendon to be evaluated on a limited number of slices (see Fig. 59.9B and C).⁴⁶ The lacertus fibrosus (bicipital aponeurosis) may prevent proximal tendon retraction after a complete distal rupture.²⁷ Partial tears of the biceps tendon are usually the result of repetitive microtrauma and tendon degeneration. Thickening of the distal biceps tendon is often seen on T1-weighted and proton density, with increased intratendinous signal intensity representing tendinopathy on T2-weighted sequences. The interval between the individual short- and long-head tendons should not be mistaken for a tear.⁴⁷ Partial tears may also produce bone marrow edema at the radial tuberosity or increased fluid in the bicipitoradial bursa on T2-weighted images.^{33,48}

US can diagnose both complete and partial tendon tears of the distal biceps. Full-thickness tears are seen as an absence of the distal biceps tendon fibers, often with interposed hematoma and adjacent edema. Partial tears at the radiobicipital origin may be difficult to diagnose on US, given the steep obliquity at the insertion of the bicipital tuberosity; however, sonographic techniques such as using a pronator window approach can improve visualization.⁴⁹

ULNAR NEUROPATHY

Ulnar neuropathy at the cubital tunnel is the most common neuropathy at the elbow. It affects 40% of athletes with valgus elbow instability and 60% of throwers with medial epicondylitis.⁵⁰ Possible causes of ulnar neuropathy potentially revealed by conventional radiographs include impinging osteophytes, heterotopic ossification, fracture malunion, and valgus instability.⁵¹ Although the history and physical examination, in conjunction with electrodiagnostic studies, are the mainstays of diagnosis, MRI may be used adjunctively to evaluate recalcitrant cases when nonoperative management or surgical intervention has failed. Vucic et al.⁵² have

suggested that MRI may actually be more sensitive than nerve conduction studies in diagnosing ulnar neuropathy at the elbow. Britz et al.⁵³ compared the efficacy of electrodiagnostic studies and MRI for the diagnosis of ulnar neuropathy. Although the two modalities were comparable for severe cases, these investigators found that when results were stratified by neuropathic severity, sensitivity for the detection of mild cases was 91% and 74% for MRI and electrodiagnostic studies, respectively. MRI findings in compression neuropathy may include constricting anatomy, nerve thickening, nerve edema, or an adjacent space-occupying lesion on T1-weighted sequences (Fig. 59.10A).⁷ In the absence of a morphologic case for cubital tunnel syndrome, it may be difficult to confidently diagnose nerve compression. The ulnar nerve commonly demonstrates increased T2-weighted signal as it passes through the cubital tunnel and has been reported in 60% asymptomatic individuals.⁵⁴ A reticulated appearance of the adjacent soft tissues or a surrounding fluid collection may represent inflammatory changes on T2-weighted sequences (see Fig. 59.10B). Chronic neuropathy with muscle denervation may produce muscle atrophy and fatty infiltration seen on T1-weighted images or muscle edema seen on T2-weighted images.⁵⁵ US is useful to assess for dynamic nerve instability or presence of accessory anconeus epitrochlearis musculature.⁵⁶

CONCLUSION

A thorough understanding of elbow pathology and a familiarity with common imaging findings ensures timely diagnosis, appropriate treatment, and expeditious return to sport for the affected athlete.

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For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

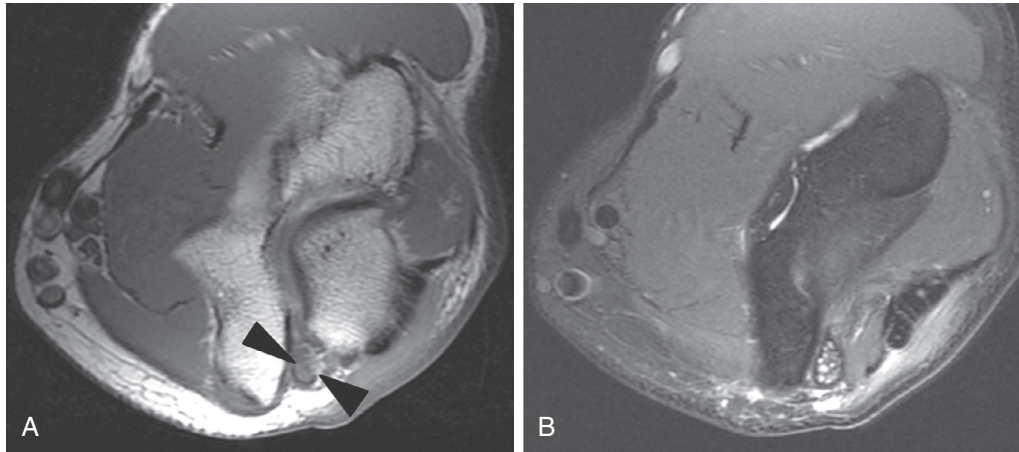


Fig. 59.10 Ulnar neuritis at the cubital tunnel. (A) Axial T1-weighted magnetic resonance imaging at the cubital tunnel demonstrates compression of the ulnar nerve with reduction of the perineural fat (arrowheads). (B) Axial T2-weighted image proximal to the cubital tunnel demonstrates prominent nerve enlargement and hyperintensity of the nerve with fascicular enlargement consistent with nerve edema/inflammation.

SELECTED READINGS

Citation:

Chen AL, Youm T, Ong BC, et al. Imaging of the elbow in the overhead throwing athlete. *Am J Sports Med.* 2003;31(3):466–473.

Level of Evidence:

III

Summary:

This article provides an excellent overview of elbow pathology and imaging in the overhead-throwing athlete.

Citation:

Norell HG. Roentgenologic visualization of the extracapsular fat; its importance in the diagnosis of traumatic injuries to the elbow. *Acta Radiol.* 1954;42(3):205–210.

Level of Evidence:

III

Summary:

In this retrospective case series, radiographs of 156 elbow fractures were examined. This article includes the original description of the posterior fat pad sign.

Citation:

Ring D, Jupiter JB, Zilberfarb J. Posterior dislocation of the elbow with fractures of the radial head and coronoid. *J Bone Joint Surg Am.* 2002;84(4):547–551.

Level of Evidence:

IV

Summary:

This article describes a clinical series of 11 patients with injuries to the radial head, coronoid, and/or lateral collateral ligament after posterior elbow dislocation. The article emphasizes the importance of recognizing the “terrible triad” and the associated role of advanced imaging.

Citation:

Zubler V, Saupe N, Jost B, et al. Elbow stiffness: effectiveness of conventional radiography and CT to explain osseous causes. *AJR Am J Roentgenol.* 2010;194(6):W515–W520.

Level of Evidence:

III

Summary:

This article describes a retrospective study of computed tomography (CT) and plain radiograph findings in 94 consecutive patients with elbow stiffness. The authors found CT helpful for the diagnosis of loose bodies. The study population was different from that in the study by Dubberley et al. (see the next citation).

Citation:

Dubberley JH, Faber KJ, Patterson SD, et al. The detection of loose bodies in the elbow: The value of MRI and CT arthrography. *J Bone Joint Surg Br.* 2005;87(5):684–686.

Level of Evidence:

III

Summary:

In this retrospective cohort study, magnetic resonance imaging, computed tomography, and conventional radiographic findings were compared with arthroscopic findings for 26 patients with mechanical symptoms of the elbow. The authors found that advanced imaging provided no benefits. Note the differences between this study and the study by Zubler et al. (see the previous citation).

Citation:

Timmerman LA, Schwartz ML, Andrews JR. Preoperative evaluation of the ulnar collateral ligament by magnetic resonance imaging and computed tomography arthrography. Evaluation in 25 baseball players with surgical confirmation. *Am J Sports Med.* 1994;22(1):26–32.

Level of Evidence:

II

Summary:

In this prospective cohort study, use of computed tomography and magnetic resonance imaging for diagnosis of medial-sided elbow pain in baseball players was evaluated. The article describes the “T sign” in partial ulnar collateral ligament tears.

Citation:

O’Driscoll SW, Bell DE, Morrey BF. Posterolateral rotatory instability of the elbow. *J Bone Joint Surg Am.* 1991;73(3):440–446.

Level of Evidence:

IV

Summary:

This article provides the first clinical description of posterolateral rotatory instability of the elbow and the role of the lateral ulnar collateral ligament in maintaining elbow stability.

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